

Krishna Koushik Unnam

AI/ML Engineer

(510)516-3735 | krishnakoushikunnam1@gmail.com | Menlo Park, CA | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Summary

AI/ML Engineer with 5+ years of experience designing, training, and deploying large-scale AI systems, including foundation models, multimodal LLMs, and enterprise machine learning solutions. Experienced in distributed model training, Generative AI, RAG, RLHF, MLOps, and scalable inference across GPU clusters and cloud platforms, delivering production-ready AI solutions for millions of users. Passionate about building intelligent systems that combine cutting-edge research with robust engineering to solve complex real-world problems at scale.

Professional Experience

Meta

Feb 2025 – Present

AI/ML Engineer

Menlo Park, CA

- Led distributed multimodal AI foundation model training pipelines using Python, PyTorch, Hugging Face Transformers, PyTorch Distributed, NCCL, and Megatron-LM, processing trillion-token datasets across multi-node GPU clusters and improving reasoning and coding benchmark accuracy by 18%.
- optimized Mixture-of-Experts (MoE) architectures using CUDA, DeepSpeed, FSDP, Megatron-LM, Tensor Parallelism, Pipeline Parallelism, and Expert Parallelism, reducing distributed training time by 32% while maximizing GPU utilization and maintaining model quality.
- Contributed scalable LLM alignment pipelines using Supervised Fine-Tuning (SFT), RLHF, Reward Models, DPO, and preference optimization in PyTorch, improving instruction following, response safety, and human evaluation scores by 21% across production models.
- Engineered large-scale data pipelines using Python, Apache Spark, Ray, Airflow, Apache Iceberg, and Parquet to automate ingestion, deduplication, preprocessing, and quality validation of petabyte-scale multimodal and synthetic training datasets.
- Developed enterprise AI Retrieval-Augmented Generation (RAG) solutions using transformer embeddings, FAISS, dense retrieval, hybrid search, and vector indexing, improving contextual reasoning and enterprise knowledge retrieval accuracy by 24%.
- Built automated AI model evaluation frameworks integrating MMLU, GPQA, HumanEval, MT-Bench, custom benchmark suites, and experiment reporting, accelerating validation cycles and ensuring deployment readiness for large foundation models.
- Containerized AI inference services using Docker, Kubernetes, gRPC, REST APIs, vLLM, TensorRT-LLM, continuous batching, KV-cache optimization, and model quantization, reducing inference costs by 35% while supporting hundreds of millions of daily inference requests.
- Implemented enterprise MLOps and LLMOps platforms using MLflow, GitHub Actions, ArgoCD, Prometheus, Grafana, and OpenTelemetry, enabling automated CI/CD, experiment tracking, model monitoring, and governance across hyperscale GPU infrastructure.
- Developed cloud-native Agentic AI platforms using Azure OpenAI Service, LangGraph, Semantic Kernel, Kubernetes, RAG, workflow orchestration, tool calling, and distributed microservices to deliver scalable enterprise AI applications.
- Collaborated with AI researchers, applied scientists, infrastructure engineers, and cross-functional product teams to productionize foundation models while optimizing model quality, scalability, inference latency, and operational efficiency across global AI platforms.

Accenture

Mar 2021 – Jun 2024

AI/ML Engineer

India

- Contributed to production-scale AI predictive maintenance pipelines using Python, Pandas, NumPy, and streaming IoT sensor data from over 15,000 industrial assets, enabling proactive failure detection and reducing unplanned equipment downtime by 32%.
- Designed and deployed machine learning models using Python, TensorFlow, Scikit-learn, XGBoost, and historical maintenance datasets to predict equipment failures, improving prediction accuracy by 21% while supporting operational decisions for over 5,000 enterprise users.
- Optimized real-time AI inference pipelines through advanced feature engineering, hyperparameter tuning, model optimization, and MLflow experiment tracking, reducing prediction latency by 38% and lowering cloud inference costs by 27% annually.
- Engineered scalable data processing pipelines using Python, Apache Spark, Databricks, SQL, Delta Lake, and ETL workflows to ingest, transform, and process billions of manufacturing sensor records for machine learning and predictive analytics.
- Built computer vision inspection systems using OpenCV, PyTorch, YOLO, and image preprocessing techniques to automate defect detection across manufacturing lines, improving product quality, inspection accuracy, and operational consistency.
- Established enterprise MLOps platforms using MLflow, Git, Docker, Kubernetes, CI/CD pipelines, model versioning, and automated monitoring, enabling reproducible model training, deployment, governance, and lifecycle management across production environments.
- Architected cloud-native AI platforms on AWS using Amazon S3, AWS Glue, SageMaker, Lambda, IAM, and CloudWatch to support secure data ingestion, feature engineering, distributed model training, deployment, and operational monitoring.
- Developed high-performance prediction microservices using Java, Spring Boot, FastAPI, REST APIs, Docker, Kubernetes, and Amazon EKS, delivering highly available real-time inference services integrated with manufacturing execution and monitoring systems.

- Engineered digital twin analytics solutions using AWS IoT Core, Databricks, streaming telemetry, and time-series analytics to simulate equipment behavior, predict failures, and optimize industrial asset performance through predictive maintenance strategies.

Technical Skills

Programming Languages: Python, SQL, Bash, Java, Go, C++, CUDA

AI/ML & Generative AI: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LLMs, Multimodal AI, Foundation Models, Mixture-of-Experts (MoE), RLHF, DPO, SFT, AI Agents, Agentic AI, LangChain, LangGraph, LlamaIndex, MCP, Multi-Agent Systems, AI Agent Orchestration, Tool Calling, Function Calling, Workflow Automation, RAG, Embedding Models, Vector Search, Computer Vision, OpenCV, YOLO, XGBoost, LoRA, PEFT, Prompt Engineering, Knowledge Distillation

Distributed Training & Model Serving: PyTorch Distributed, FSDP, DeepSpeed, Megatron-LM, Ray, NCCL, vLLM, TensorRT-LLM, Model Quantization, GPU Optimization, Distributed Inference, HPC, Tensor Parallelism, Pipeline Parallelism, Data Parallelism

Data Engineering & Big Data: Apache Spark, Databricks, Apache Airflow, Apache Kafka, ETL Pipelines, Data Processing, Feature Engineering, Data Validation, Data Quality, Synthetic Data Generation, Streaming Data Pipelines, IoT Data Processing

Cloud Platforms & Infrastructure: AWS (S3, SageMaker, Glue, Lambda, EKS, IoT Core), Microsoft Azure, Google Cloud Platform (GCP), Spring Boot, Kubernetes, Docker, REST APIs, Microservices, Service Mesh, Infrastructure as Code (IaC), Terraform

MLOps, DevOps, Testing & Monitoring: MLflow, Git, GitHub Actions, CI/CD, ArgoCD, Pytest, Model Deployment, Model Versioning, Model Monitoring, Experiment Tracking, Prometheus, Grafana, OpenTelemetry, Kubernetes Operations, Feature Stores, AI Application Deployment

Evaluation, Databases, Vector Stores & Analytics: MMLU, GPQA, HumanEval, Custom Model Benchmarking, A/B Testing, PostgreSQL, MySQL, pgvector, Pinecone, Redis, FAISS, Power BI, Performance Optimization, Explainable AI (XAI), LangSmith

Professional Skills: System Design, Distributed Systems, Cross-Functional Collaboration, Technical Leadership, Analytical Thinking, Agile Development, Mentoring, Project Ownership, Documentation, Debugging, Innovation, Time Management, Continuous Learning

PROJECTS

HELIX — PRODUCTION LLM INFERENCE GATEWAY | [GitHub](#)

Go, PostgreSQL, pgvector, Redis, Docker, Prometheus, Grafana, React

- Architected and deployed a multi-provider LLM inference gateway with score-based provider routing, pgvector semantic caching, JWT authentication with tenant-scoped controls, Redis token-bucket rate limiting, circuit-breaker fallback, SSE streaming, structured request logging, and Prometheus/Grafana observability.
- Load-tested the live, cache-warmed gateway across 3,885 requests and up to 100 concurrent virtual users, achieving 0% request failures, 18.43 requests per second, 567 ms p95 latency, and a 99.97% semantic-cache hit rate.

MULTI-AGENT CODE REPAIR PLATFORM | [GitHub](#)

Python, LangGraph, LiteLLM, GitHub API, tree-sitter, Docker, Pytest

- Built a stateful issue-to-pull-request workflow with repository research, minimal-fix planning, commit-pinned patch generation, Docker-isolated validation, structured retry feedback, and human escalation for ambiguous or unsuccessful repairs.
- Validated the workflow on two real open-source Click issues, passing 1,136–1,436 tests with clean Ruff and Mypy checks, zero retries, approximately two-minute execution time, and \$0.35–\$0.36 model cost per run.

DOCUMIND — EXPLAINABLE ENTERPRISE RAG PLATFORM | [GitHub](#)

Python, FastAPI, React, Pinecone, LangChain, LlamaIndex, OpenAI API

- Built an end-to-end document intelligence platform that extracts and chunks PDF content, generates embeddings, stores document-scoped vectors in Pinecone, and serves context-grounded answers through an asynchronous FastAPI backend and React interface.
- Implemented document- and chunk-level metadata filtering, retrieved-source evidence, explicit empty-retrieval handling, and LlamaIndex relevancy evaluation to reduce cross-document contamination and make generated responses traceable to supporting context.

Education

Master of Science in Computer Science & Engineering

University of South Florida

Certifications

AWS Certified Machine Learning Engineer – Associate (MLA-C01)

AWS Certified AI Practitioner

Microsoft Certified: Azure AI Engineer Associate

Databricks Certified Machine Learning Professional